Latent-Function Video Diffusion: Generate Once, Query Videos at Any Continuous Control

Anonymous CVPR submission

Paper ID *****

Abstract

*Continuously controlled video generators are usually queried pointwise, so every new control value requires another denoising trajectory. We propose **Latent-Function Video Diffusion (LFVD)**, a family-valued formulation in which one trajectory samples a reusable latent representation and a lightweight continuous decoder reads complete videos at arbitrary scalar controls without further transformer evaluations. LFVD combines a family encoder, a query decoder, standard flow matching, and a frozen video VAE; its encoder can be direct identity packing, learned temporal–control mixing, or a compact map. On five object-disjoint OpenIllumination folds, the default learned-mixing instance improves LPIPS from .4000 to .3216, PSNR from 19.29 to 21.14 dB, and control correlation from .786 to .933 over shared-noise Pointwise while using one trajectory rather than nine. Trained at three measured lights and evaluated on six omitted positions of the locked test objects, it improves LPIPS from .4103 to .3410 and PSNR by 2.41 dB; correlation rises from .604 to .991. Synchronized timing and analytical transformer FLOPs show amortization after approximately seven Lighting requests. An encoder ablation shows that identity packing matches learned mixing overall and is strongest on four actor-disjoint MEAD folds, where LFVD preserves Pointwise fidelity with one rather than three trajectories but does not improve ordering over endpoint interpolation. The gains therefore support family-valued generation rather than superiority of a particular encoder.*

1. Introduction

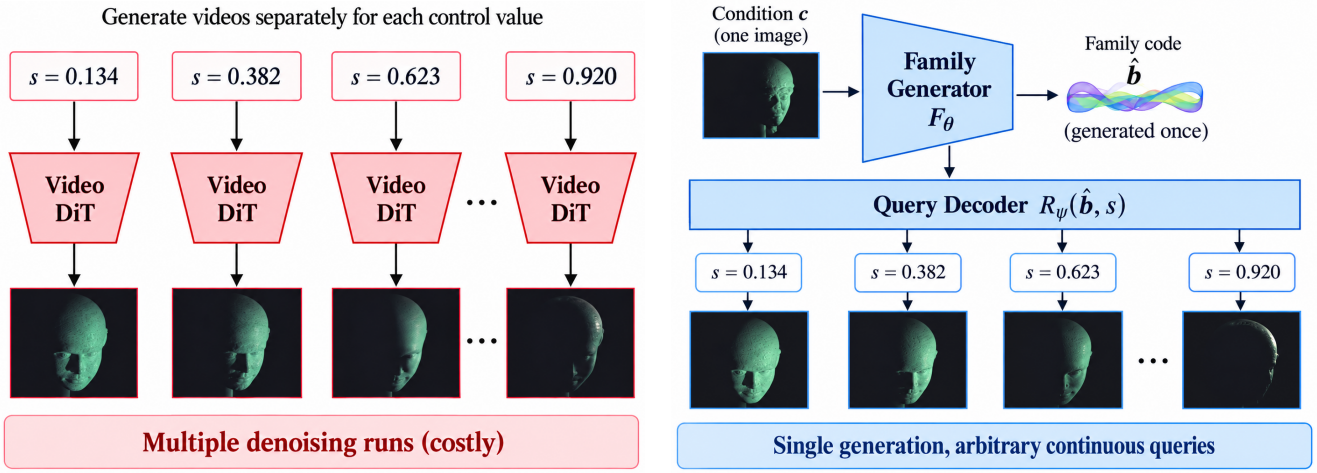
Recent video diffusion models have made remarkable progress in visual quality and controllability. As these models mature, an emerging goal is to generate not just a single video, but a structured family of related videos under continuous control. Modern video generators synthesize appearance, temporal dynamics, and diverse content from text or image conditions [2, 11, 12, 24]. Users may wish to vary the strength of an expression while keeping the same speaker, or change illumination while preserving an object and camera route. These requests share most of their content and differ along a control axis. In interactive workflows the user often explores many nearby values before selecting an output. Repeated generation then spends most of its computation reconstructing content that is expected to remain consistent. This motivates learning the control-dependent family as a shared generative object.

Our conventional Pointwise sampler accepts a control value s during sampling and produces the corresponding video. Changing s requires another denoising trajectory. Reusing the same initial noise couples the outputs, but still requires evaluating the denoiser for each requested value. Generating a small number of endpoint videos and interpolating their latents is another practical alternative. Both are important comparisons: the first preserves per-value generation, and the second avoids most repeated sampling by fixing an interpolation rule.

A natural alternative is to generate the whole family jointly. Packing all observed control videos into one sequence and attaching a continuous learned reader already yields a queryable family from one trajectory. We show that this is not a competing formulation but the identity-encoder instance of *family-valued diffusion*: one sampled latent object followed by arbitrarily many post-sampling queries. LFVD makes the family encoder an explicit design choice and tests which parts of the formulation produce the observed gains.

We represent the complete family of control-dependent videos with one shared latent object. A continuous query selects a complete video latent after generation, so its coordinate is a control value across videos rather than a frame timestamp. The query accepts any scalar $s \in [0, 1]$; its fidelity at unobserved controls is tested using real measured targets.

Specifically, we introduce **Latent-Function Video Diffusion (LFVD)**. A family encoder combines the measured control videos into one latent tensor. A jointly trained query



(a) Pointwise: separate diffusion for each control.

(b) LFVD: one generated code, multiple queries.

Figure 1. **Generate once, query many videos.** Conventional per-control sampling repeats the Video DiT trajectory for every requested s. LFVD samples a single family code and evaluates a learned decoder for further queries with no additional DiT sampling. Thumbnails are archived held-out OpenIllumination frames; cost trade-offs are measured in Sec. 4.5.

decoder reconstructs individual video latents from that tensor and their control values. A video diffusion transformer then learns the distribution of these family representations. At inference time, the family encoder and observed target videos are not needed: the diffusion model samples a family code from the input condition, and the query decoder reuses it for every requested value. The query is learned directly, with no prescribed interpolation basis, generated-anchor sorting, or additional denoising. Figure 1 illustrates this separation schematically.

LFVD shares one generative trajectory across requests, but its family latent is longer than an ordinary video latent. We therefore report synchronized time, analytical transformer FLOPs, and compact-encoder trade-offs rather than infer efficiency from trajectory counts alone.

We study two control domains. OpenIllumination supplies measured photographs of the same object under different lights, arranged along a common camera route. MEAD supplies native facial videos at three expression-intensity levels. On held-out objects, LFVD improves lighting LPIPS from 0.4000 to 0.3216 and PSNR from 19.29 to 21.14 dB against shared-noise Pointwise. On held-out actors, LFVD obtains competitive reconstruction quality, although endpoint interpolation retains a stronger ordering score. These results separate the ability to reuse a generated family from the quality of its individual outputs and its response to control. We further train on only three measured lights and test the six omitted physical positions across all five object folds. This is the direct continuous-query test: no evaluation target is synthesized by interpolation. Finally, synchronized timing and a dense-matmul FLOP analysis report the actual multi-query crossover rather than inferring efficiency from trajectory counts. An encoder ablation further shows that direct identity packing matches learned mixing overall and is stronger on MEAD fidelity, locating the principal benefit in family-valued generation rather than a particular encoder.

Contributions.

- We formulate continuous video control as family-valued diffusion: one sampled latent object followed by arbitrarily many post-sampling scalar queries.
- We develop LFVD on a pretrained Video DiT: a family encoder, a direct scalar-conditioned query decoder trained by measured VAE-latent reconstruction, and standard flow matching on the family code. Identity packing, learned temporal–control mixing, and compact encoders are instances of the same framework.
- We provide object- and actor-disjoint evaluation, a five-fold test on genuinely unseen physical controls, encoder ablations on both datasets, paired cluster bootstrap intervals, and synchronized timing plus analytical FLOPs that locate the crossover against batched Pointwise.

2. Related Work

2.1. Video Diffusion Models

Diffusion models extend image synthesis to jointly model spatial appearance and temporal dynamics. Video Diffusion Models [12], Make-A-Video [24], and Imagen Video [11] developed spatiotemporal denoising, transferred image priors, and cascaded video synthesis. Latent-space approaches improve scalability by operating on compressed video representations. Video LDM [2] adds temporal layers to pretrained image latent diffusion models, while Projected

Video Diffusion Models [30] learn compact video representations. Lumiere [1] models the full temporal extent through a space–time architecture. More recent video backbones, including Wan [27], provide pretrained video autoencoders and transformers that can be adapted to downstream generation tasks. These advances supply the visual representations and generative priors used by controllable video systems.

2.2. Motion and Intensity Control in Video Generation

Explicit motion conditions extend video control beyond text and reference images. DragNUWA [29], Motion-I2V [22], and Motion Prompting [9] use trajectories or motion fields, while MotionPro [33] adds region-aware control. Go-with-the-Flow [3] injects motion through warped noise, and TokenMotion [13] separates human and camera controls. MotionStone [21] and Mojito [10] address motion magnitude. Text Slider [4], FreeSliders [7], and TokenDial [18] provide continuous attribute controls. These methods apply control within generation; shared-prefix approaches can reuse part of a trajectory before control-specific denoising. In a typical conditional sampler, the selected control is supplied during denoising. Shared-prefix methods can reduce repeated work but may still require control-specific sampling suffixes. LFVD evaluates new control values with a learned decoder after the family trajectory is complete. The comparison is therefore about reusable sampled state, not merely the precision of an individual motion controller.

2.3. Joint Families and Multi-Illumination Data

Joint multi-view diffusion models such as MVDream [23] and SyncDreamer [17] generate fixed view sets in one reverse process. Analogously, packing multiple control slots into one sequence is a direct family encoder. Within LFVD, raw packing with a learned continuous reader is exactly the identity-encoder instance; comparison with learned encoders separates the family-valued formulation from encoder choice. For controlled lighting, OpenIllumination [16] provides real OLAT measurements of the same object under many lights. We use it as a measured continuous-control benchmark, not as evidence for a lighting-specific model: the same family codec, flow objective, and query rule are used for facial expression.

2.4. Functional Generative Modeling and Efficient Sampling

Functional generative models learn distributions over coordinate-dependent signals. Neural Diffusion Processes [5], Score-Based Diffusion Models in Function Space [14], Continuous-Time Functional Diffusion Processes [8], and Functional Diffusion [31] study function-valued representations and generation. HyperDiffusion [6] generates neural-field weights. These works establish that continuously queryable generative representations are not unique to video control. Sampling acceleration addresses a complementary question: DDIM [25], progressive distillation [20], Consistency Models [26], and Latent Consistency Models [19] reduce evaluations along a sampling trajectory. LFVD learns a representation whose external query coordinate selects an entire video within a control family, allowing one completed trajectory to serve multiple requests. This external coordinate differs from a spatial or temporal coordinate inside one generated signal: every queried point is an entire spatiotemporal latent. The proposed training pipeline combines a learned family codec with flow-based conditional sampling of its latent code. Its novelty claim is consequently narrower than "diffusion over functions": we study how a pretrained Video DiT can sample a reusable family of complete videos, when that representation is useful, and how its encoder affects the quality–compute trade-off.

3. Latent-Function Video Diffusion

LFVD separates family representation, generation, and query. It is defined by one latent object sampled with one diffusion trajectory and reused for every control value, not by a particular encoder. Fixed, learned, full-length, and compact encoders are all admissible. Figure 2 shows the two training stages and inference path.

3.1. Conditional Videos and Latent Functions

Let c denote the shared input condition, including a reference image and text context, and let $s \in \mathcal{S} = [0, 1]$ be a continuous control value. A video x_s has latent $z_s = \mathcal{E}(x_s) \in \mathcal{L}$, where $\mathcal{E}$ and $\mathcal{D}$ are the frozen video VAE encoder and decoder, and $\mathcal{L}$ is the video-latent space. The Pointwise sampler, with parameters ϑ and initial noise ξ, generates

$$\hat{z}_s = G_\vartheta(c, s, \xi), \qquad \hat{x}_s = \mathcal{D}(\hat{z}_s). \tag{1}$$

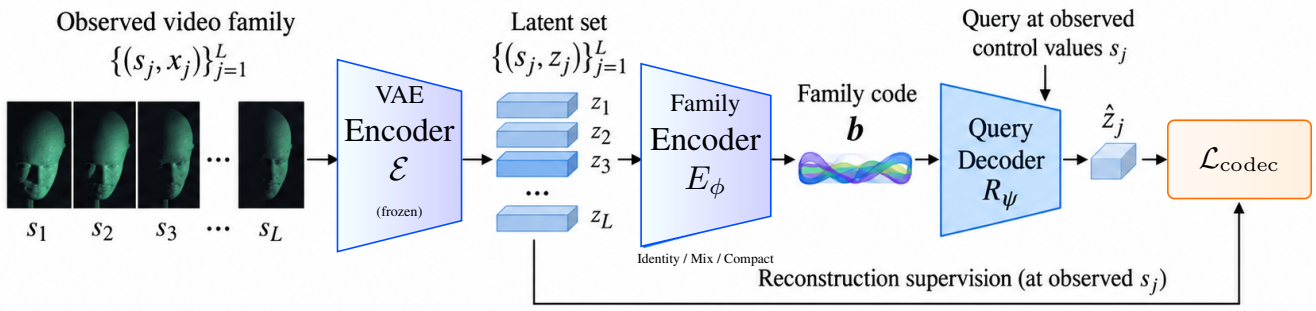
Each new s requires another evaluation of the complete sampler. LFVD represents the control-dependent mapping with a family code $\mathbf{b}$ and a learned query decoder R_ψ:

$$\mathcal{Z}_\mathbf{b} : \mathcal{S} \to \mathcal{L}, \qquad \mathcal{Z}_\mathbf{b}(s) = R_\psi(\mathbf{b}, s). \tag{2}$$

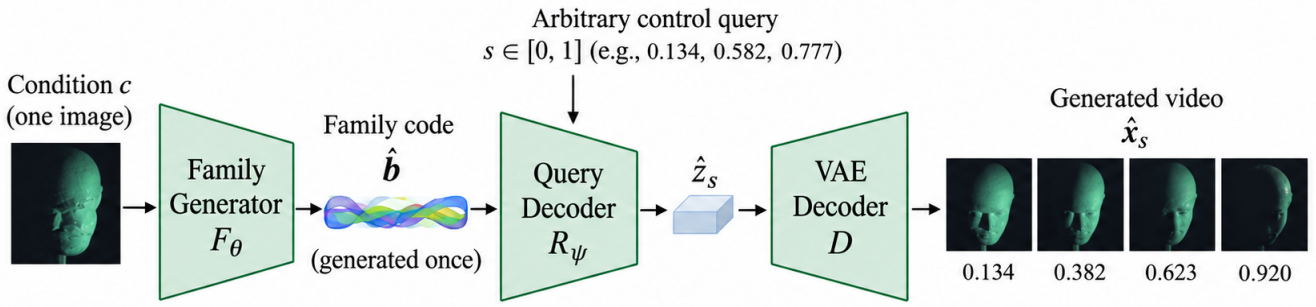
A single sampled code therefore defines all queried outputs, independently of its encoder parameterization. The formulation does not impose semantic monotonicity or guarantee accuracy at controls absent from training.

3.2. Learning a Family Representation

For each condition c, training provides L measured videos $\mathcal{F}_c = \{(s_\ell, x_{s_\ell})\}_{\ell=1}^{L}$. Their VAE latents have C channels, T_z latent frames, and spatial size $H \times W$. Let r denote the observed reference latent shared across controls, let z_ℓ^+ denote the remaining $T_z - 1$ latent frames, and set $J =$



(a) Family codec learning at observed controls.



(b) Single family-code generation followed by direct continuous queries.

Figure 2. **LFVD training and inference.** (a) The frozen VAE $\mathcal{E}$ and family codec (E_ϕ, R_ψ) are trained only with $\mathcal{L}_{\text{codec}}$. The frozen codec then supplies targets for separate $\mathcal{L}_{\text{FM}}$ Video DiT training; the losses are never summed. (b) R_ψ answers arbitrary queries without another DiT trajectory. Identity, Mix, and Compact differ only in E_ϕ.

$L(T_z - 1)$. A family encoder E_ϕ, whose parameters may be empty for a fixed encoder, maps the complete family to one state,

$$\mathbf{b}_0 = E_\phi(\{(s_\ell, z_\ell)\}_{\ell=1}^{L}; r) = [r, \mathbf{b}_0^+]. \tag{3}$$

Brackets concatenate along latent time. The encoder chooses $\mathbf{b}_0^+ \in \mathbb{R}^{C\times J'\times H\times W}$. LFVD requires one code per family and one shared reader, not a particular J' or learned mixer.

Encoder instances. Our evaluated encoders are covered by one temporal–control map. We order the observations by their known coordinates and optionally add an embedding $g_\phi(s) \in \mathbb{R}^C$, broadcast over space and the $T_z - 1$ frames. With embedding scale λ, define

$$\begin{aligned} U &= \mathrm{Concat}_{\ell,\mathrm{time}}\big(z_\ell^+ + \lambda g_\phi(s_\ell)\big), \\ \mathbf{b}_0^+ &= P_\phi U, \qquad P_\phi \in \mathbb{R}^{J'\times J}. \end{aligned} \tag{4}$$

The same equation gives *LFVD-Identity* ($J' = J$, $\lambda = 0$, $P_\phi = I$), *LFVD-Mix* ($J' = J$ with learned square P_ϕ and g_ϕ), and *LFVD-Compact* ($J' < J$ with rectangular P_ϕ). Mix initializes P_ϕ to identity and uses a bounded two-layer SiLU embedding with $\lambda = 0.25$. Other encoders are valid if they produce one queryable code.

Direct continuous query. A scalar-conditioned hypernetwork produces a temporal map $W_\psi(s) \in \mathbb{R}^{(T_z-1)\times J'}$ and a channel–time bias $\beta_\psi(s) \in \mathbb{R}^{C\times(T_z-1)}$. It reconstructs the requested suffix and restores the shared reference:

$$\begin{aligned} \hat{z}_s^+ &= W_\psi(s)\mathbf{b}_0^+ + \beta_\psi(s), \\ R_\psi(\mathbf{b}_0, s) &= [r, \hat{z}_s^+]. \end{aligned} \tag{5}$$

The temporal map is applied at every spatial location and channel; the bias is broadcast spatially. Its weights depend on s and are learned with the encoder. No fixed basis, anchor-recovery step, or analytic interpolation rule is prescribed. The encoder and decoder are trained only with queried VAE-latent reconstruction,

$$\mathcal{L}_{\text{codec}} = \mathbb{E}_{\mathcal{F}_c}\left[\frac{1}{L}\sum_{\ell=1}^{L} \|R_\psi(\mathbf{b}_0, s_\ell) - z_\ell\|_1\right], \tag{6}$$

where the norm is averaged over latent elements. The VAE remains frozen; there is no perceptual, ordering, or sharpening objective.

Shared semantics across encoders. All instances share the codec objective, reader, Video DiT rule, and inference. On a fixed grid, reader parameters can absorb the coordinate offset and an invertible square P_ϕ; Identity and Mix thus

have equal nominal decodable capacity but different diffusion coordinates. Compact also trades length for fidelity. Section 4.4 tests these effects.

3.3. Diffusion over the Family Code

After codec training, E_ϕ and R_ψ are frozen. Each training family is encoded once using Eq. 3, and a Video DiT learns the conditional distribution of its raw code. We retain the pretrained backbone's flow-matching objective [15]. For noise level $\sigma \in [0,1]$ and Gaussian noise ε of the same shape as the generated suffix, the interpolation and velocity target are

$$\begin{aligned} \mathbf{b}_\sigma &= [r, (1-\sigma)\mathbf{b}_0^+ + \sigma\varepsilon], \\ u^\star &= \varepsilon - \mathbf{b}_0^+. \end{aligned} \tag{7}$$

The observed reference is not a generation target. Writing v_θ^+ for the predicted suffix velocity, training minimizes

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}\left[w(\sigma)\|v_\theta^+(\mathbf{b}_\sigma, \sigma, c) - u^\star\|_2^2\right], \tag{8}$$

with the upstream scheduler's timestep sampling and weighting w. The same velocity-learning principle is used for Pointwise. LFVD changes the target representation, and does not add a teacher, residual target, or sampled-code repair.

At inference, the flow sampler starts from $[r, \xi]$, with Gaussian suffix ξ, and integrates from the noise end toward the data end while keeping r fixed. The complete generation path is

$$\begin{aligned} \hat{\mathbf{b}} &= F_\theta(c, \xi), \\ \hat{x}_s &= \mathcal{D}\Big(R_\psi(\hat{\mathbf{b}}, s)\Big). \end{aligned} \tag{9}$$

The requested s enters only the query decoder. The family encoder and all target videos are absent from this inference path. Every query returns a complete video, and the Video DiT is not called again.

3.4. Representation Size and Generation Cost

For an encoder output length J', the family code has shape $C \times (1 + J') \times H \times W$ per example, or five dimensions including the batch axis. Dense Identity and Mix use $J' = J$. With $C = 48$, $T_z = 5$, and $H = W = 16$, their shapes are $48 \times 37 \times 16 \times 16$ for nine lighting controls and $48 \times 13 \times 16 \times 16$ for three MEAD controls; Compact uses a smaller J' selected before diffusion training. Each query returns $48 \times 5 \times 16 \times 16$, decoded to a 17-frame 256×256 video. We retain these dimensions throughout the main study. The sparse Lighting Identity/Mix models trained at three controls also use 13 latent frames, 64.9% fewer than the 37-frame dense code but still 2.6× an ordinary five-frame video latent. Across the locked five-fold test it retains near-dense all-control fidelity (.3271 versus .3216 LPIPS) while moving the analytical Pointwise crossover to 2.37 requested videos. This crossover is a FLOP result, not a sparse-code wall-clock measurement.

Let M be the number of requested outputs. Define $\mathcal{C}_{\mathrm{diff}}^{\mathrm{video}}$ and $\mathcal{C}_{\mathrm{diff}}^{\mathrm{family}}$ as the costs of sampling one ordinary video latent and one family latent, respectively; let $\mathcal{C}_R$ and $\mathcal{C}_D$ be the costs of a query and a VAE decode. Sequential generation costs are

$$\begin{aligned} \mathcal{C}_{\mathrm{base}}(M) &= M(\mathcal{C}_{\mathrm{diff}}^{\mathrm{video}} + \mathcal{C}_D), \\ \mathcal{C}_{\mathrm{LFVD}}(M) &= \mathcal{C}_{\mathrm{diff}}^{\mathrm{family}} + M(\mathcal{C}_R + \mathcal{C}_D). \end{aligned} \tag{10}$$

Shared conditioning work is omitted from both expressions. The family trajectory is longer and need not have the cost of one ordinary trajectory. For a fixed trained representation, its denoising cost does not depend on M, but querying and decoding do. Trajectory and call counts therefore describe reuse, not an equivalent wall-clock or FLOP reduction.

4. Experiments

We evaluate three questions. (1) Can one sampled family code match or improve pointwise quality and control? (2) Which benefits come from family-valued generation, and which from the family encoder? (3) At how many post-hoc queries does its larger trajectory amortize? All principal results use held-out objects or actors, three fixed sampling seeds, 50 denoising steps, and the official trainer of the pretrained backbone.

4.1. Datasets and Protocol

OpenIllumination. OpenIllumination [16] provides real objects captured under one-light-at-a-time illumination. We form a video family by fixing an object and 17-frame camera path while varying only light position. Nine measured illuminations follow their physical angular order and are normalized by arc distance to $s \in [0,1]$; no evaluation target is interpolated. Five object-disjoint folds contain 48/16/16 train/validation/test families and cover 80 held-out families in total. The same object never appears across splits.

MEAD. MEAD [28] contains talking-face recordings at three ordinal emotion intensities. We use measured low/mid/high videos and construct 17-frame families after MFCC+delta dynamic-time-warp alignment, so corresponding clips describe the same speech phase rather than arbitrary timestamps. Four actor-disjoint folds each use five training actors, one validation actor, and two test actors; together they contain 112 held-out families. This is a native video benchmark, but its labels are ordinal rather than a calibrated physical axis.

Human-subject data. MEAD contains identifiable faces and voices. We use the pre-existing research dataset under

its terms and collect no new personal data or participant interactions; consequently, this work conducted no new IRB review or consent process.

Matched training. Both methods use the Wan2.2-TI2V-5B Video DiT and frozen video VAE [27] through the official DiffSynth trainer. Videos are RGB, 256×256, and 17 frames. We train the family codec for 8,000 steps, then freeze it and train rank-64 LoRA adapters on attention and MLP projections for 3,456 updates. Pointwise uses the same backbone, LoRA rank, optimizer, updates, split, and conditioning. Sampling uses 50 steps and seeds $\{100, 200, 300\}$. No experiment uses a teacher, residual target, perceptual loss, ordering loss, or query-time DiT.

Baselines and encoder instances. *Pointwise independent* runs one trajectory per value with independent noise. *Pointwise shared-noise* reuses initial noise but still runs one complete trajectory per value. *Endpoint interpolation* samples the two endpoint videos and linearly interpolates their VAE latents. Unless stated otherwise, LFVD denotes LFVD-Mix. LFVD-Identity directly packs all observed controls and uses the same learned reader; LFVD-Compact shortens the family code.

Metrics and statistics. We report AlexNet LPIPS [32], PSNR, and SSIM against official measured targets. Control correlation is Spearman correlation between requested s and a target-projected response statistic; a constant output receives zero. Endpoint amplitude compares generated and target response ranges, and path second variation measures smoothness across queries. Confidence intervals are paired cluster bootstraps over complete held-out objects or actors, never individual frames. Full settings and per-sample records are in the supplement.

4.2. Held-Out Main Results

Table 1 shows a clear Lighting result: relative to the strong shared-noise baseline, LFVD improves LPIPS by .0784, PSNR by 1.85 dB, and control correlation by .146 while replacing nine trajectories with one. The object-bootstrap intervals for LFVD are [.3096,.3344] LPIPS, [20.58,21.65] dB PSNR, and [.905,.960] correlation. Independent noise severely weakens ordering, confirming that shared noise is the appropriate Pointwise comparator.

MEAD gives a more qualified result. LFVD matches shared-noise Pointwise fidelity (.1151 versus .1157 LPIPS) and has greater endpoint response, but its correlation is .719 versus .743, while two-endpoint interpolation reaches .859. The result supports amortized family generation in a dynamic domain, not a claim that LFVD improves every ordinal control path. We retain this neutral result because it exposes the difference between physically ordered lighting and coarsely labeled expression intensity.

4.3. Unseen Controls

The main protocol evaluates measured controls used during training. To test the functional claim directly, we retrain Pointwise and LFVD using only light indices $\{0, 4, 8\}$ and evaluate all nine official images. The other six positions are genuine held-out controls. We repeat the complete protocol on all five object-disjoint folds and three seeds. No intermediate target enters the codec or flow objectives. We evaluate each fold on its locked test objects; separately archived validation results agree closely.

Across 80 locked-test families, 20 objects, and three seeds, LFVD-Mix improves unseen-control LPIPS by .0693 (95% paired object-bootstrap CI $[.0576, .0825]$), PSNR by 2.41 dB ($[2.13, 2.69]$), and SSIM by .0248 ($[.0191, .0305]$). Over all nine queries, control correlation rises from .604 to .991; its paired gain is .386 ($[.316, .464]$). Thus the continuous-query result holds on official test images absent from both codec and flow training.

4.4. Encoder Ablation: Where Do the Gains Come From?

With a complete grid, LFVD-Identity is statistically unresolved from LFVD-Mix: its LPIPS difference (Identity minus Mix) is $-.0014$ with 95% paired object-bootstrap CI $[-.0058, .0030]$. The corresponding PSNR, SSIM, and correlation intervals also contain zero. The locked sparse test gives the same conclusion overall: Identity minus Mix LPIPS is $+.0007$ ($[-.0015, .0028]$) and correlation is $+.0013$ ($[-.0037, .0069]$). Only the tiny unseen PSNR difference, $-.05$ dB ($[-.10, -.002]$), resolves in favor of Mix. On MEAD the ordering reverses, with Identity improving LPIPS by .0082 (95% paired actor-bootstrap CI $[.0002, .0204]$) and PSNR by .19 dB ($[.00, .46]$). Encoder choice is therefore dataset-dependent; the robust result is the family-valued formulation and continuous reader.

4.5. Measured Multi-Query Cost

Figure 4 corrects the tempting but incomplete argument from trajectory counts. The family code contains more tokens than one video, and Pointwise batching matters. LFVD is not a one-query speedup. It amortizes after roughly seven requests and is $1.23\times$, $1.97\times$, and $2.90\times$ faster than batch-3 Pointwise at $M = 9, 17, 33$, respectively. At the measured crossover point $M = 7$, the totals are 15.40 versus 16.96 seconds, a $1.10\times$ speedup. On MEAD's three-query grid, LFVD takes 5.74 seconds versus 6.60 seconds for batch-3 Pointwise ($1.15\times$) and 11.47 seconds for sequential Pointwise ($2.00\times$).

Table 5 counts the patch projections, all self- and cross-attention projections and products, feed-forward linear

Table 1. **Held-out main results.** LFVD denotes the default LFVD-Mix instance. Means cover five object-disjoint Lighting folds or four actor-disjoint MEAD folds, three seeds, and 50 steps. Traj. and calls are per family. Best and second-best fidelity or ordering are bold and underlined within each dataset.

Dataset	Method	LPIPS↓	PSNR↑	SSIM↑	ρ ↑	Traj.	DiT calls
Lighting	Pointwise independent	.3979	19.69	.737	.468	9	450
	Pointwise shared-noise	.4000	19.29	.731	.786	9	450
	Endpoint interpolation	.4059	19.33	.727	.860	2	100
	LFVD	**.3216**	**21.14**	**.753**	**.933**	**1**	**50**
MEAD	Pointwise independent	**.1063**	**22.97**	**.764**	.577	3	150
	Pointwise shared-noise	.1157	22.54	.753	.743	3	150
	Endpoint interpolation	.1152	22.91	.762	**.859**	2	100
	LFVD	.1151	**22.97**	.760	.719	**1**	**50**

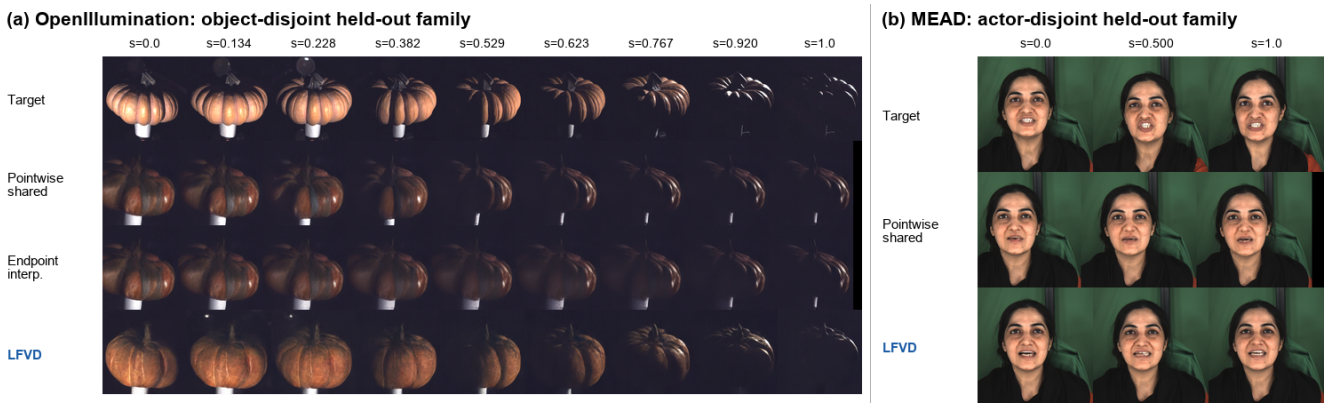


Figure 3. **Actual held-out continua at a fixed video frame.** OpenIllumination uses all nine measured light positions; MEAD uses three measured intensity levels. Each row comes from one generated family and is shown without sharpening or spatial alignment. Lighting exhibits a strong ordered response. The selected MEAD family illustrates the harder regime: fidelity is preserved, but the ordinal response is visibly weaker.

Table 2. **Sparse-control Lighting, five folds.** Observed uses the three training controls; unseen uses six official intermediate illuminations. Final aggregation was locked to 50 steps and three seeds before evaluation.

Subset	Method	LPIPS↓	PSNR↑	SSIM↑
Observed	Pointwise shared	.4062	18.48	.7181
	LFVD-Mix	**.2993**	**20.99**	**.7532**
Unseen	Pointwise shared	.4103	18.81	.7232
	LFVD-Mix	**.3410**	**21.22**	**.7480**

Table 3. **Encoder ablation.** LFVD-Identity and LFVD-Mix share the reader, Video DiT, LoRA, updates, and 50-step, three-seed sampling. Full and sparse results use five object-disjoint locked test folds. Sparse image metrics use six unseen measured positions; ρ uses all nine queries.

Subset	Method	LPIPS↓	PSNR↑	SSIM↑	ρ ↑
Full	LFVD-Identity	**.3202**	**21.14**	.751	.929
	LFVD-Mix	.3216	**21.14**	**.753**	**.933**
Sparse-unseen	LFVD-Identity	.3417	21.17	.747	**.992**
	LFVD-Mix	**.3410**	**21.22**	**.748**	.991

Table 4. Measured seconds versus requested videos M; three repeats.

Method	1	3	5	7	9	17	33
Pointwise sequential	3.88	11.45	19.06	–	35.38	64.67	127.50
Pointwise batch-3	**3.90**	**6.62**	**11.16**	16.96	19.86	37.65	72.84
LFVD	13.19	13.93	14.66	**15.40**	**16.15**	**19.14**	**25.11**

maps, and text projection, with one multiply–accumulate counted as two FLOPs. It gives hardware-independent crossovers of 2.37 requested videos for MEAD and 6.82 for Lighting. Thus the measured Lighting crossover is not merely a 24-GB batching artifact: at $M = 7$, Pointwise requires 1,214.47 TFLOPs versus 1,183.59 for LFVD, and at $M = 9$ the ratio is 1.32×.

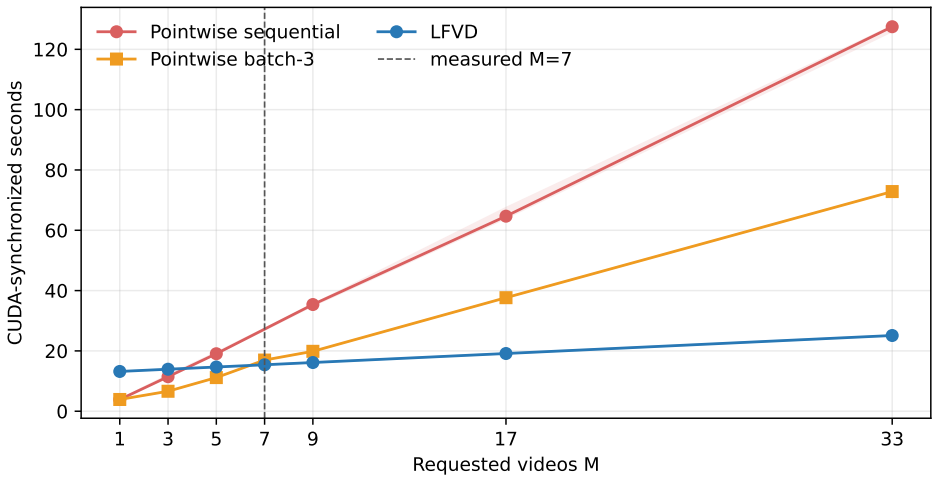


Figure 4. **Synchronized 50-step wall-clock scaling on one RTX 4090.** Each point averages three resident-model repeats and includes querying and VAE decoding, but excludes model loading and file serialization. Pointwise batch-3 is the strongest memory-feasible baseline. LFVD is slower for few queries and crosses the fitted batched curve at 6.64 queries; the measured $M = 7$ point confirms the crossover rather than extrapolating through it.

Table 5. Hardware-independent dense-matmul compute for the locked 50-step Wan2.2-TI2V-5B transformer. Query decoding and VAE decoding are excluded and reported instead by the synchronized timing experiment.

Representation	Latent frames	Tokens	TFLOPs/trajectory
Pointwise video	5	320	173.50
MEAD family	13	832	411.52
Lighting family	37	2,368	1,183.59

Table 6. **Where quality is lost.** Exact-code rows bypass diffusion. Rows 1–3 use the 25-step validation ablation; compact sampling uses the locked L0 record, 50 steps, and three seeds. Sampled rows use one rank-64 trajectory.

Representation	LPIPS↓	PSNR↑	SSIM↑	$\rho\uparrow$
Frozen VAE target	.0512	33.17	.899	–
Exact family code	.0642	32.06	.893	1.000
Sampled family code	.3365	20.86	.749	.973
Compact exact (28 frames)	.0814	30.84	.884	1.000
Compact sampled (28 frames)	.4053	21.06	.725	.977

4.6. Ablations and Bottleneck

The exact-code oracle nearly matches the frozen VAE and has perfect ordering, so the learned coordinate and query decoder can represent the family. The remaining error comes mainly from sampling the larger code manifold. Increasing LoRA rank from 16 to 64 improves sampled LPIPS from .3610 to .3365 and correlation from .925 to .973. Increasing training families from 1 to 48 monotonically improves LPIPS from .450 to .337 and correlation from .411 to .973, while exact-code quality remains approximately .064 LPIPS. These trends argue against single-family memorization and identify generative modeling of the family code as the current bottleneck.

LFVD-Compact can shorten the code: the 28-frame variant removes 24.3% of family tokens while retaining .4053 LPIPS and .977 correlation after sampling, but is worse than the matched full-length .3300 LPIPS. Under the same dense-matmul accounting, this shifts the analytical Pointwise crossover from 6.82 to 5.09 requested videos; no compact-code wall-clock claim is made.

4.7. Limitations

LFVD trades trajectories for a latent of size $1 + L(T-1)$ and is useful only beyond the measured crossover. OpenIllumination is controlled multi-view data; MEAD is native video but has only three ordinal levels, where LFVD does not beat endpoint interpolation in ordering. Encoder differences are small and dataset-dependent, so we claim the family-valued formulation rather than one encoder. Claims about encoder choice are limited to Identity, Mix, and Compact. Generated quality also remains below the exact-code oracle, leaving family-code diffusion as the clearest target for future work.

5. Conclusion

LFVD samples a reusable family object for post-hoc queries. It improves Lighting quality and control, generalizes to locked-test intermediate positions, and amortizes over enough requests; MEAD preserves fidelity, not superior ordering. Identity matches Mix overall, locating the contribution in family-valued generation.

LLM Usage Statement

OpenAI Codex was used as an engineering and writing assistant for research ideation, implementation and debugging of experiment code, experiment automation, preparation of figures and tables, numerical-consistency checks, and prose editing. The human authors defined the research questions and methodology, selected the data and evaluation protocols, inspected the generated outputs, verified the reported results and citations, and retain full responsibility for the submission.

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